CM3020 Artificial intelligence

- Lecture 1
- We will start in 10 mins
- Please download slides “Artificial Intelligence_lecture 1” from SIM Canvas

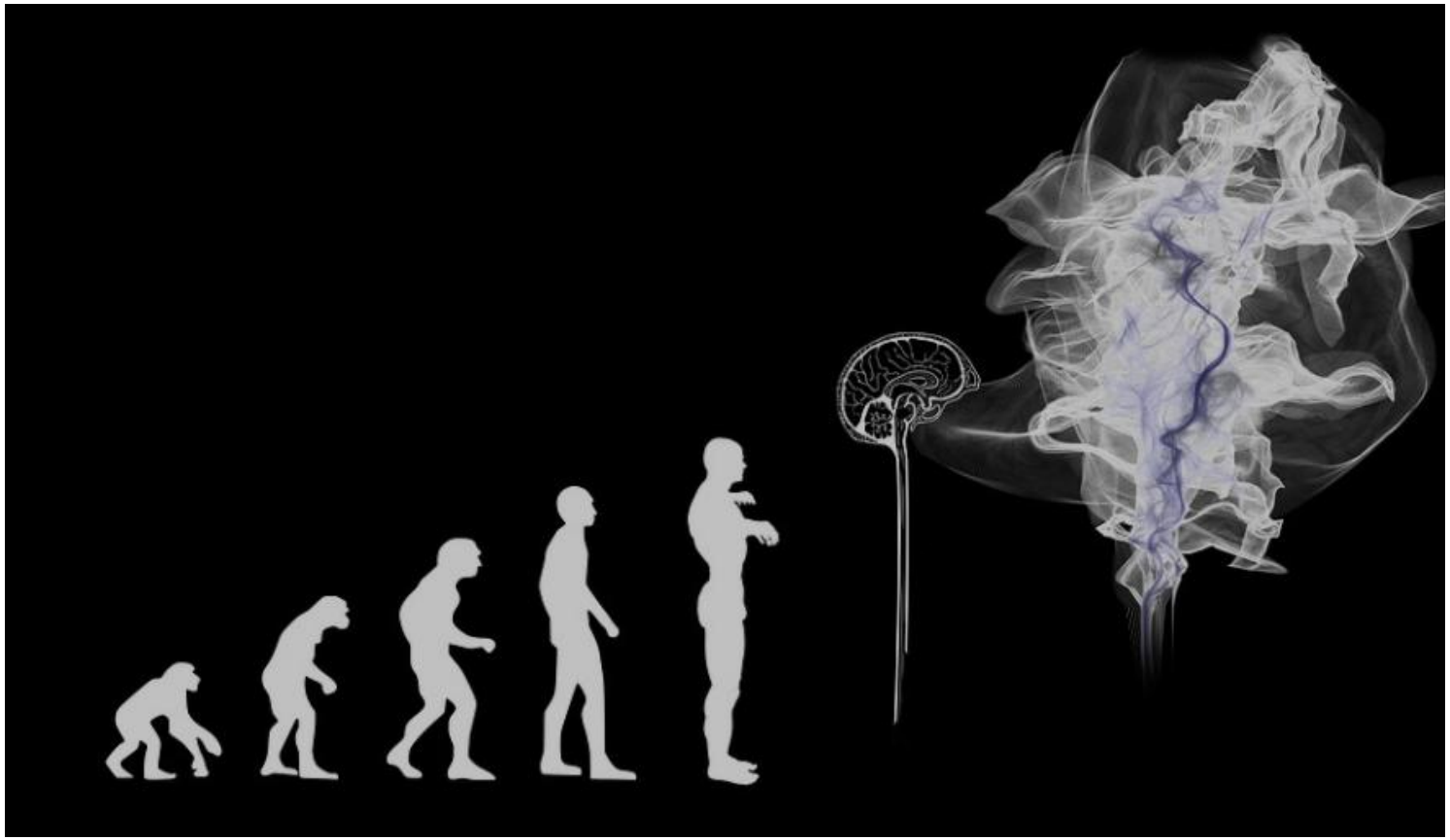


Administrative details

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- Email: dckpeh001@mymail.sim.edu.sg
- In class lecture slides may contain more examples and answers



Introduction

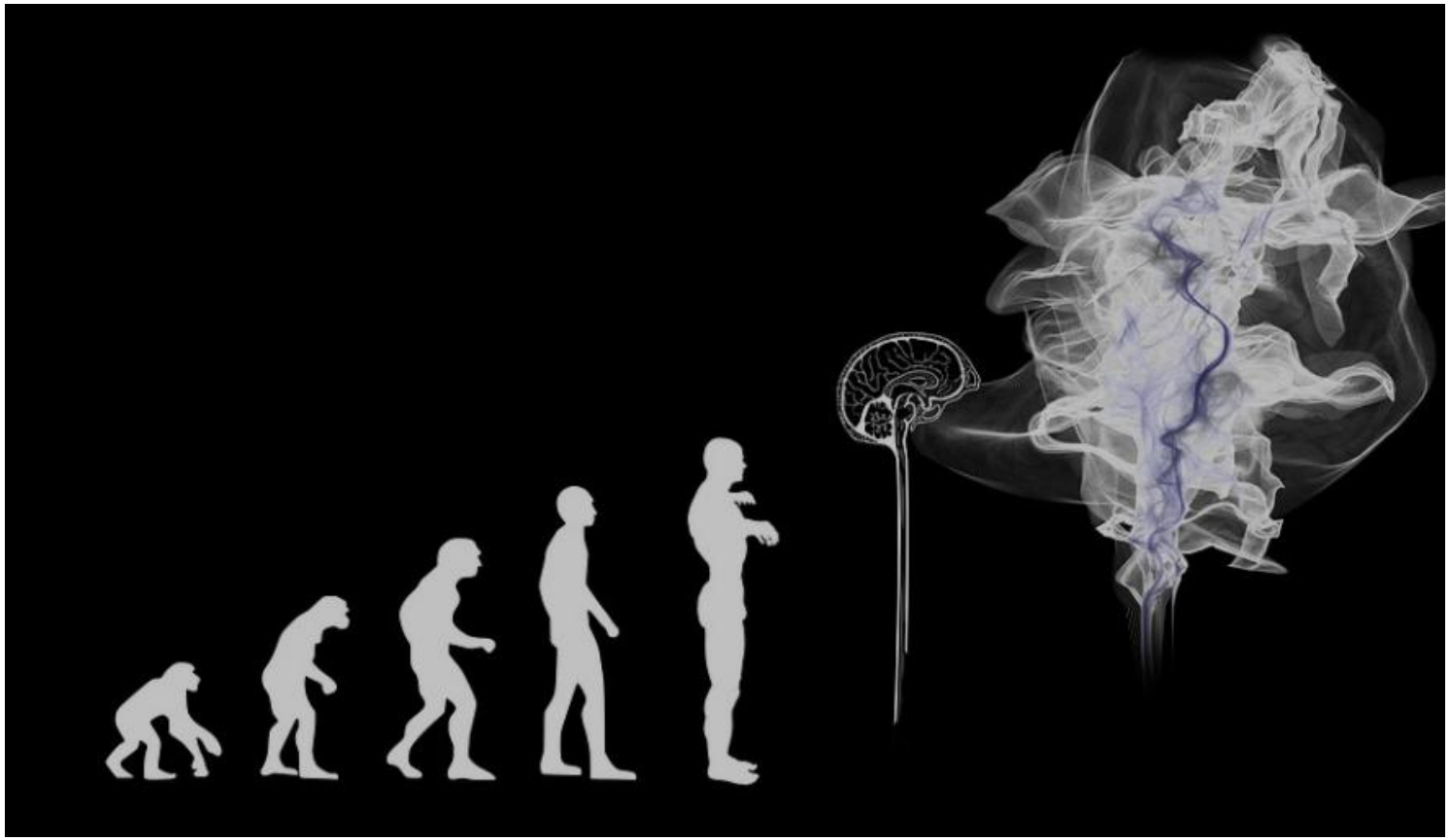


For the first lecture ONLY, we will divert away from the direction in Coursera

Because I want to introduce you to the breadth of the amazing world of

AI

Introduction



What do you think is Artificial Intelligence ?

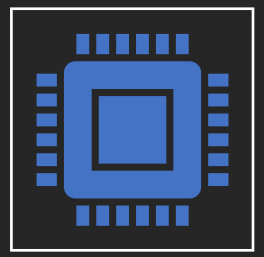
Introduction

Thinking Humanly	Thinking Rationally
Acting Humanly	Acting Rationally

- What is the difference between **acting** and **thinking**?
- Behavior

- What is the difference between **humanly** and **rationally**?
- Thought processes and reasoning

Introduction



Thinking Humanly

' ... effort to make computers think ... ' (Haugeland, 1985)

'[The automation of] activities that we associate with human thinking ...' (Bellman, 1978)



The cognitive modelling approach

Can we “copy and paste” a human mind into a machine?

We must then first understand the human mind

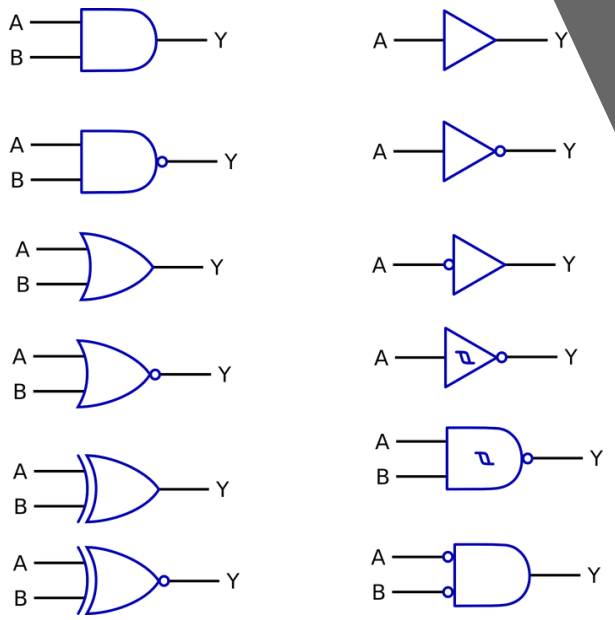
Introspection: trying to catch our own thoughts as they go by

psychological experiments: observing a person in action; and through

brain imaging: observing the brain in action



Introduction





Thinking Rationally

'The study of mental faculties through the use of computer models.' (Charniak and McDermott, 1985)

'The study of the computations that make it possible to perceive, reason, and act.' (Winston, 1992)



The laws of thought approach

All modules are boring. CM3020 is a module. CM3020 is boring?

A field known as logic

Can we write or express everything in a language?

Theory vs reality: Do we have sufficient computing resources?

Introduction



- Acting Rationally
  - ' ... the study of the design of intelligence agents.' (Poole et al., 1998)
  - ' ... intelligence behaviour in artifacts.' (Nilsson, 1998)

- The **rational agent** approach
  - To achieve the best (expected) outcome, given uncertainty
  - Is gambling profitable?
  - Careful deliberation vs reflex

Introduction



- Acting Humanly
  - ' ... creating machines that perform functions that require intelligence when performed by people.' (Kurzweil, 1990)’
  - ... how to make computers do things at which, at the moment, people are better.' (Rich and Knight, 1991)

- The **Turing Test** approach
  - Covered briefly later and in depth in later lectures

History of AI

- AI is one of the newest fields in science and engineering
- Work started soon after World War II, and the name itself was coined in 1956
- Truly a universal field
  - huge variety of subfields, ranging from the general (learning
  - and perception) to the specific, such as playing chess, proving mathematical theorems, writing
  - poetry, driving a car on a crowded street, and diagnosing diseases

History

- 1950: Paper "Programming a Computer for Playing Chess“
  - A chess playing algorithm by Claude Shannon - https://en.wikipedia.org/wiki/Shannon_number
  - Hand implemented by Alan Turing - https://www.history.com/news/in-1950-alan-turing-created-a-chess-computer-program-that-prefigured-a-i
- 1952: First checkers by Arthur Samuel wrote a checkers program, a program that he would spend two decades developing and refining. Written in assembly language, Samuel’s program used both alpha-beta searching and learning.
  - https://www.theatlantic.com/technology/archive/2017/07/marion-tinsley-checkers/534111/
- 1979: Backgammon
  - https://bkgm.com/articles/Berliner/BackgammonProgramBeatsWorldChamp/
- 1989: Computer Olympiad (computer versus computer competition)
  - https://en.wikipedia.org/wiki/Computer_Olympiad
- 1992: a world championship caliber checkers playing game system Schaeffer’s
  - https://www.sciencedirect.com/science/article/abs/pii/0004370292900748
  - https://www.theatlantic.com/technology/archive/2017/07/marion-tinsley-checkers/534111/
- 1997: The first match was played in Philadelphia in 1996 and won by Kasparov by 4–2. A rematch was played in New York City in 1997 and won by Deep Blue by 3½–2½.
  - https://en.wikipedia.org/wiki/Deep_Blue_versus_Garry_Kasparov
- 2002: World championship player standard Scrabble AIs
  - http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.2.9747&rep=rep1&type=pdf

History

- 2007: Completely solved checkers, plays the best possible game every time. A weak solution showing checkers always results in a draw if neither player makes a mistake.
  - https://en.wikipedia.org/wiki/Jonathan_Schaeffer
- 2009: First Mario AI competition
  - https://www.researchgate.net/publication/224177833_The_2009_Mario_AI_Competition
- 2015: Atari
- 2018: Texas Hold'em Poker
- 2019: Dota 2 AI, called OpenAI Five
  - https://openai.com/five/
- 2020: Atari 57 & Agent 57, the first deep reinforcement learning agent to obtain a score that is above the human baseline on all 57 Atari 2600 games
  - https://www.deepmind.com/blog/agent57-outperforming-the-human-atari-benchmark#:~:text=human%20Atari%20benchmark-,March%2031%2C%202020,all%2057%20Atari%202600%20games.
- https://academicworks.cuny.edu/cgi/viewcontent.cgi?article=1181&context=gc_pubs

Goal of AI



- To understand and replicate human thought processes through computational modelling
  - Moved to Cognitive Science
- Alan Turing's famous 'Turing test' for intelligence
  - Proposed in 1950
  - designed to provide a satisfactory operational definition of intelligence.
  - a program is judged intelligent if its behaviour cannot be differentiated from that of a human
- Ex Machina
  - https://www.youtube.com/watch?v=d3LIXhE1ReE&ab_channel=hollywoodstreams

Cognitive Science: Inter-disciplinary field (AI, psychology, linguistics, philosophy, anthropology) that tries to form computational theories of human cognition

Turing's Test



Acting humanly

If the human cannot tell whether the responses from the other side of a wall are coming from a human or computer, then the computer is intelligent.



Current State of AI

- Most modern AI programs are instead designed to **act rationally**
  - Take the best possible action given their **goals**, **knowledge** and **constraints**
- Why design to act rationally?
  - It provides a concrete, objective measure against which performance can be judged and understood
  - Not required to model poorly understood areas of human's odd habit /unusual feature
  - Able to build systems that may be useful in situations in which humans do not always behave rationally
- Would you dare to take a ride in a Tesla that acts humanly?

Subfields of Artificial Intelligence

The major subfields of Artificial Intelligence	Problem solving
	Knowledge representation and reasoning
	Planning
	Learning
	Vision
	Natural language

Subfields of Artificial Intelligence - Problem solving

An agent is given a problem setting and a goal and must determine how to realize that goal

Example

Chess	Sudoku	Tower of Hanoi

Find a sequence of operations to produce the expected output from the initial input

Subfields of Artificial Intelligence - Knowledge representation and reasoning studies

- How an agent can represent knowledge it has about the environment
- Uses it to derive further knowledge,
- Either using
  - a logic-based representation (when the knowledge is certain) or
  - a probabilistic representation
- Example
  - Self driving cars
  - Perception -> **Knowledge representation -> reasoning** -> Planning -> Execution

Knowledge representation and reasoning

AI agents deal with knowledge (data)

- Facts (believe & observe knowledge)
- Procedures (how to process knowledge)
- Meaning (relate & define knowledge)

Right representation is crucial

- Wrong choice can lead to project failure

Examples

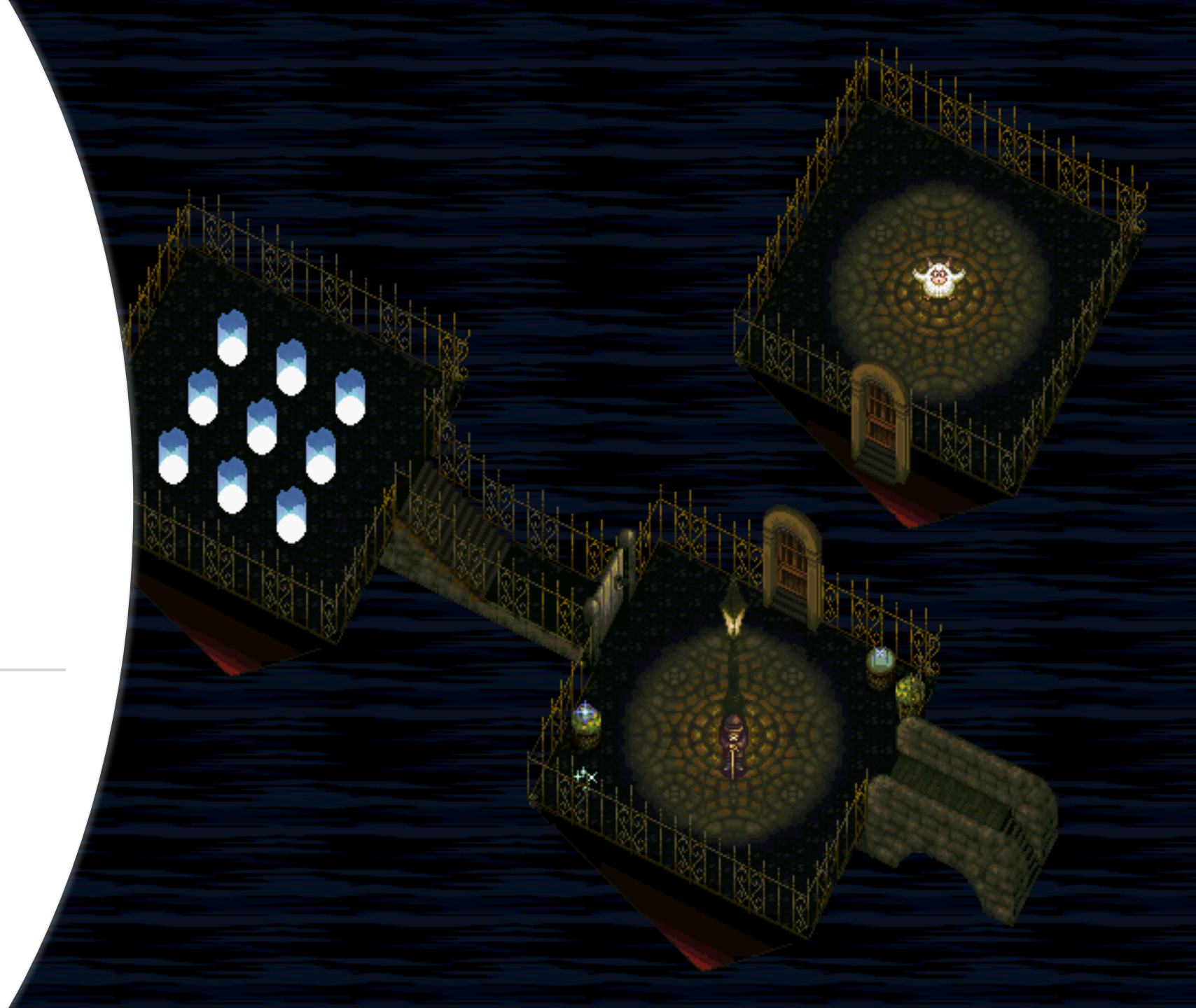
- First order theorem proving… first order logic
- Inductive logic programming… logic programs
- Neural networks learning… neural networks

Subfields of Artificial Intelligence - Planning

- An agent is given **knowledge of an environment** and **formulates a plan** for interacting with it to **achieve its goals**.
- What is planning?
  - A collection of actions for performing some task
  - Example: assembling your new IKEA cupboard.
  - Programs are available to help humans formulate plans.
  - But it is difficult to generate plans automatically.
- Example
  - Self driving cars
    - Perception -> Knowledge representation -> Reasoning -> **Planning** -> Execution

Break

We will be back in **10 mins**



Subfields of Artificial Intelligence - Learning

- An agent improves its performance through experience.
  - ability to distinguish between categories of objects (supervised learning),
  - learning structure from raw data (unsupervised learning),
  - learning to maximise reward (or minimise cost) in an environment (reinforcement learning)



Subfields of Artificial Intelligence - Learning

- Learning denotes changes in a system that will enable the system to do the same task more efficiently next time.
  - Example: email spam detection system
    - Trained on email messages to learn to distinguish between spam and non-spam messages
    - After learning, it can be used to classify new email messages into spam and non-spam folders

Subfields of Artificial Intelligence - Vision

- Computer Vision is the science and technology of obtaining models, meaning and control information from visual data
- An agent interprets or processes raw visual images
  - How about videos?
  - Acquiring, analysing, processing and understanding images and videos
  - The Robocup tournament and ASIMO are examples of Artificial Intelligence using Computer Vision to its greatest extent.



Subfields of Artificial Intelligence - Natural language Processing

- Branch of computer science focused on developing systems that allow computers to communicate with people using everyday language
- An agent must process input in a natural language (e.g. English),or generate it.
  - Example: Siri
  - Challenges involving natural language understanding
    - How to enable computers to derive meaning from human or natural language input
  - Can music be a natural language?
    - https://youtu.be/UWxfnNXlVy8?t=240



Turing's Test







Acting humanly

If the human cannot tell whether the responses from the other side of a wall are coming from a human or computer, then the computer is intelligent.

Tying it back to Turing Test

- The computer would need to possess the following capabilities:
  - Natural language processing to enable it to communicate successfully in English
  - Knowledge representation and reasoning
    - Knowledge representation to store what it knows or hears
    - Automated reasoning to use the stored information to answer questions and to draw new conclusions
  - Machine learning to adapt to new circumstances and to detect and extrapolate patterns







Current state of the Turing Test

- https://www.zdnet.com/article/the-turing-test-has-a-problem-and-openais-gpt-4-5-just-exposed-it/
- https://arxiv.org/abs/2503.23674

Moravec paradox

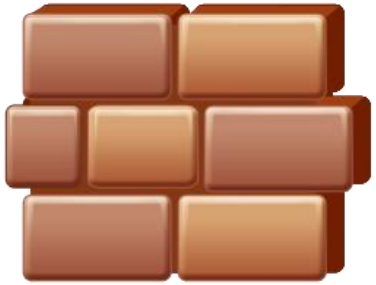
- Reasoning requires very little computation
  - Logic and algebra are difficult for people and are considered a sign of intelligence.
- Expectation: the "easy" problems of vision and commonsense reasoning would soon fall into place
  - BUT Sensorimotor and perception skills require enormous computational resources
- Then comes AI winter (~1965), a period of reduced funding and interest in artificial intelligence research



Total Turing test

- Turing's test deliberately avoided direct physical interaction between the interrogator and the computer, because physical simulation of a person is unnecessary for intelligence.
- A modified version, to includes a video signal so that
  - the interrogator can test the subject's **perceptual abilities**
  - the opportunity for the interrogator to pass physical objects "through the hatch."
- To pass the total Turing test, the computer will need
  - computer vision to perceive objects, and
  - robotics to manipulate objects and move about







Current State of AI

- https://www.instagram.com/p/DLARun3xwUV/
- https://www.instagram.com/p/DLq3mKYzKBC/

Introduction

- Why we want AI playing games
  - To build a basis for Strong / General AI. Then move onto solving real world issues
    - Test and train artificial intelligence in a low-stakes environment
    - Testbeds for developing new AI techniques which paves the way for innovations in other fields
    - Watson (NLP & Chess) became the basis for different domains including healthcare, cybersecurity, and weather forecasting
  - Commercial interest
    - AI to play with us!
    - Play testing
  - Game AI means useful metrics
    - offer a way to measure the progress and ability of AI

Introduction

Philosophy of AI: AI can be broken into two distinct areas (Camp)

Weak AI	**Strong AI**
• AKA Narrow AI or artificial narrow intelligence (ANI) • Computer programs can be built that **acts** intelligently • solve problems or handle tasks that would typically be considered to require intelligence • Disregards if a computer can think or not	• AKA broad AI, general AI or artificial general intelligence (AGI) • Machine that behaves like a human has a mind and is capable of **actually thinking** • Not just a **simulation** of thinking
• Weak AI **appears** to be a human mind • Performs the single task for which it is trained for • Can be strong in one area and weak in others • AlphaGo • https://deepmind.com/research/case-studies/alphago-the-story-so-far	• Strong AI **is** a human mind • Think and act intelligently like humans • Close(?) but not yet: ChatGTP 4.0?

Introduction

Super AI is AI that surpasses human intelligence and ability. It's also known as artificial superintelligence (ASI) or superintelligence. It's the best at everything — maths, science, medicine, hobbies, you name it. Even the brightest human minds cannot come close to the abilities of super AI.

- Extract from https://www.thinkautomation.com/bots-and-ai/types-of-ai-distinguishing-between-weak-strong-and-super-ai/ [THIS IS OUT OF COURSE GUIDE SCOPE]

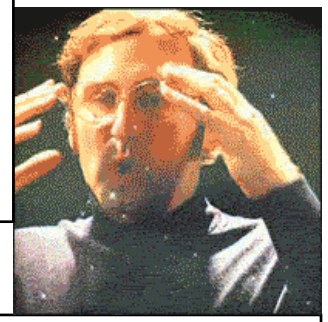
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The evolution

- https://g.co/gemini/share/12ba98d587d7

AI playgame

- https://youtu.be/DcYLT37ImBY?si=UUrk5DtQga7l-cOy
  - Playing Pokemon via reinforcement learning
- https://youtu.be/NfGcWGaO1E4
  - 25 ChatGPT AIs Play A Game - So What Happened?